Componentwise polymatroidality is not preserved by homological shifts

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Abstract

We give a negative answer to a question of Ficarra on homological shifts of componentwise polymatroidal ideals. Over any field, the ideal $(x_1, \ldots, x_r, yz, y^a)$ is componentwise polymatroidal for every $r \geq 1$ and $a \geq 3$, but each of its intermediate homological shift ideals fails polymatroidality in every sufficiently large homogeneous component. A tensor product of two elementary resolutions gives all the shifts, and a single exchange obstruction proves the failure uniformly. The smallest example also contradicts the first-shift assertion in Theorem 10.1 of Ficarra–Lu, arXiv:2509.11977v1; we identify the invalid step in its proof.

1 Statement and context

Let R be a polynomial ring over a field K, with its standard grading. For a monomial ideal J, write $J_{\langle d\rangle}$ for the ideal generated by its degree-d monomials. The ideal J is *componentwise polymatroidal* if each nonzero $J_{\langle d\rangle}$ is polymatroidal. Recall that a monomial ideal generated in one degree is polymatroidal if, whenever u, v are minimal generators and $\deg_t u > \deg_t v$, there is a variable s with $\deg_s u < \deg_s v$ such that su/t is a generator.

The ith homological shift ideal $\mathrm{HS}_i(J)$ is generated by the monomials corresponding to the multigraded shifts in homological position i of a minimal multigraded free resolution of J. Thus $\mathrm{HS}_0(J) = J$; the indexing refers to the ideal, rather than its quotient.

Ficarra [2, Question 9] asks whether every homological shift of a componentwise polymatroidal ideal is again componentwise polymatroidal. For equigenerated polymatroidal ideals, preservation of polymatroidality has been proved by Cid-Ruiz, Matherne, and Shapiro [1, Theorem A]. Ficarra–Lu [3, Theorem 10.1] assert the first-shift case of the componentwise question. The following family answers the question negatively and shows that this latter assertion, as stated in the cited version, is false.

Theorem 1. *Let K be any field, $r \geq 1$, and $a \geq 3$. In*

$$R = K[x_1, \ldots, x_r, y, z], \qquad I = (x_1, \ldots, x_r, yz, y^a),$$

the ideal I is componentwise polymatroidal and $\mathrm{pd}_R I = r + 1$. For every $1 \leq i \leq r$ and every $d \geq a + i$, the ideal $\mathrm{HS}_i(I)_{\langle d\rangle}$ is not polymatroidal. The top shift is the principal ideal

$$\mathrm{HS}_{r+1}(I) = (x_1 \cdots x_r y^a z).$$

In particular, failure occurs in arbitrarily many homological positions simultaneously, and it persists in all sufficiently large components of each intermediate shift. The proof does not depend on the characteristic of K.

2 Proof

Put $X = \{x_1, \ldots, x_r\}$. For an integer j, let $E_j(X)$ be the ideal generated by the squarefree monomials of degree j in X, with $E_0(X) = R$ and $E_j(X) = 0$ for $j < 0$ or $j > r$.

First we verify the hypothesis of Theorem 1. The degree-one component of I is (X). For $2 \le d < a$, a degree-d monomial is in I precisely when it involves a variable of X or both y and z. Therefore

$$\mathcal{G}(I_{\langle d\rangle}) = \{u : \deg u = d,\ \deg_y u \le d-1,\ \deg_z u \le d-1\}.$$

For $d \ge a$, the only degree-d monomial outside I is z^d, so

$$\mathcal{G}(I_{\langle d\rangle}) = \{u : \deg u = d,\ \deg_z u \le d-1\}.$$

Both descriptions are bounded Veronese sets and satisfy exchange directly: if $\deg_t u > \deg_t v$, equality of total degrees supplies a variable s with $\deg_s u < \deg_s v$, and increasing $\deg_s u$ by one respects every upper bound since $\deg_s u + 1 \le \deg_s v$. Thus I is componentwise polymatroidal.

Proposition 2. *For every $i \ge 0$,*

$$\mathrm{HS}_i(I) = E_{i+1}(X) + yzE_i(X) + y^aE_i(X) + y^azE_{i-1}(X). \tag{1}$$

Proof. Write $A = K[X]$ and $B = K[y,z]$. There is an isomorphism

$$R/I \cong A/(X) \otimes_K B/(yz, y^a).$$

The first factor has its minimal Koszul resolution, with squarefree x-multidegrees of degree j in position j. The second has the minimal multigraded resolution

$$0 \longrightarrow B(-y^az) \xrightarrow{\binom{y^{a-1}}{-z}} B(-yz) \oplus B(-y^a) \xrightarrow{(yz\quad y^a)} B \longrightarrow B/(yz, y^a) \longrightarrow 0.$$

The notation $B(-u)$ records the multidegree of the monomial u. Exactness follows because a relation $fyz + gy^a = 0$ gives $fz = -gy^{a-1}$, and the coprime monomials z, y^{a-1} divide g, f, respectively. All matrix entries have positive degree. Tensoring these resolutions over the field K gives a minimal free resolution over R: exactness follows from the tensor product over a field, and minimality follows because all differential entries remain in the homogeneous maximal ideal. Multiplying the multidegrees and adding homological positions gives the shifts of R/I. Position $i+1$ of that resolution is position i of the resolution of I, yielding (1). □

For $i = r+1$, only the last summand of (1) survives and is nonzero; for $i > r+1$ all summands vanish. This proves the projective-dimension and top-shift assertions.

Fix $1 \le i \le r$, choose $F \subseteq X$ with $|F| = i-1$, and choose $x_t \in X \setminus F$. Write $x_F = \prod_{x\in F} x$. For $q \ge 0$, set

$$u = x_Fx_ty^{a-2}z^{q+2}, \qquad v = x_Fy^az^{q+1}.$$

These monomials have degree $d = a+i+q$ and belong to $\mathrm{HS}_i(I)$: the second summand of (1) contains u, and the fourth contains v. They are consequently minimal generators of $\mathrm{HS}_i(I)_{\langle d\rangle}$. The exponent of x_t in u exceeds that in v, and the only variable with smaller exponent in u than in v is y. Exchange would therefore require

$$w = yu/x_t = x_Fy^{a-1}z^{q+2}$$

to belong to $\mathrm{HS}_i(I)$. It does not: its x-support has size $i-1$, excluding the first three summands of (1), while its y-exponent is smaller than a, excluding the fourth. This proves Theorem 1.

3 The smallest example and the first-shift assertion

For $r = 1$ and $a = 3$, the example is

$$I = (x, yz, y^3), \qquad \mathrm{HS}_1(I) = (xyz, xy^3, y^3z).$$

Its degree-four component contains xyz^2 and y^3z but not the exchange monomial y^2z^2. Meanwhile $I_{\langle 2\rangle} = (x^2, xy, xz, yz)$, and for every $d \geq 3$ the degree-d monomials of I are all such monomials except z^d.

The proof of [3, Theorem 10.1] asserts

$$\mathrm{HS}_1(J)_{\langle j\rangle} = \mathrm{HS}_1(J_{\langle j-1\rangle}) \tag{2}$$

for a componentwise polymatroidal ideal J. This equality already fails for the principal ideal $J = (x)$ in $K[x, y]$: at $j = 3$, the left side is zero, whereas

$$\mathrm{HS}_1(J_{\langle 2\rangle}) = \mathrm{HS}_1((x^2, xy)) = (x^2y).$$

The argument for (2) passes from least common multiples of distinct generators of a homogeneous component to least common multiples of distinct minimal generators of J. The former two monomials may instead be multiples of the same minimal generator of J. Theorem 1 shows that the final first-shift assertion itself fails, in addition to this intermediate equality. These observations concern the componentwise assertion only and do not address the equigenerated or asymptotic results of [3].

References

[1] Y. Cid-Ruiz, J. P. Matherne, and A. Shapiro, *Syzygies of polymatroidal ideals*, arXiv:2507.13153v1, July 17, 2025. `https://arxiv.org/abs/2507.13153`.

[2] A. Ficarra, *Shellability of componentwise discrete polymatroids*, Electron. J. Combin. **32**(1) (2025), P1.41, published March 14, 2025. `https://www.combinatorics.org/ojs/index.php/eljc/article/view/v32i1p41`.

[3] A. Ficarra and D. Lu, *Polymatroidal ideals and their asymptotic syzygies*, arXiv:2509.11977v1, September 15, 2025. `https://arxiv.org/abs/2509.11977v1`.